

Distal Myopathy 6, Adult-Onset, Autosomal Dominant (MPD6 / ACTN2 Actininopathy)

Comprehensive Disease Characteristics Report

Summary

Distal Myopathy 6, adult-onset, autosomal dominant (MPD6; OMIM #615197; MONDO:0014138) is a rare, slowly progressive skeletal muscle disease caused by heterozygous (dominant) variants in **ACTN2** (alpha-actinin-2), the major structural protein of the sarcomeric Z-disk (HGNC:164; gene locus 1q42–1q43; UniProt P35609). Because the causative gene is *ACTN2*, the disease is often called "**actininopathy**." It was first delineated as a distinct genetic distal myopathy by Savarese and colleagues in 2019, who described four families (three Spanish, one Swedish) with adult-onset, asymmetric distal muscle weakness beginning with impaired ankle dorsiflexion (foot drop) and later spreading to proximal limb muscles [PMID: 30900782].

The core disease mechanism is a **toxic gain-of-function / dominant-negative process driven by misfolding and aggregation of mutant alpha-actinin-2 at the Z-disk**. Cell-model work shows that dominant, protein-extending frameshift variants form alpha-actinin-2 aggregates, whereas recessive missense variants do not — directly implicating aggregation as the dominant disease mechanism [PMID: 39095936]. Aggregation is accompanied by Z-line disorganization, minicore/core formation, and progressive fiber loss, culminating in distal-predominant weakness. Zebrafish and mouse models expressing mutant alpha-actinin-2 recapitulate sarcomeric disorganization, core pathology, and impaired muscle force [PMID: 30701273].

Clinically, the dominant distal/limb-girdle form is genotype-dependent, ranging from asymptomatic carriers to limb-girdle weakness with facial involvement, and — importantly — **generally spares the heart and respiratory system** [PMID: 39812845]. Muscle MRI shows a characteristic pattern (tibialis anterior, soleus, gluteus medius/minimus, hamstrings), which

now aids diagnosis. There is **no curative therapy**; management is supportive (ankle-foot orthoses, physiotherapy) with genetic counseling. This report synthesizes eight confirmed findings drawn from 30 reviewed papers into a full disease characterization spanning etiology, phenotype, genetics, mechanism, anatomy, temporal course, epidemiology, diagnostics, prognosis, treatment, prevention, and model organisms.

1. Disease Information

Overview. MPD6 is a Mendelian, adult-onset, autosomal dominant distal myopathy of skeletal muscle. It belongs to the broad category of distal myopathies — genetic primary muscle disorders that present with prominent weakness at onset in the hands and/or feet — which are increasingly recognized as a subset of the muscular dystrophies (progressive genetic diseases with loss of muscle fibers) [PMID: 33458580]. Savarese et al. established that dominant *ACTN2* variants define a new genetically determined distal myopathy, coining the term "actininopathy": "*ACTN2 encodes for alpha actinin2, which is highly expressed in the sarcomeric Z-disk with a major structural and functional role. Actininopathy is thus a new genetically determined distal myopathy.*" [PMID: 30900782].

Key identifiers.

Resource	Identifier
OMIM	#615197 (Myopathy, distal, 6, adult-onset, autosomal dominant; MPD6)
Mondo	MONDO:0014138
Gene	ACTN2 — HGNC:164; NCBI Gene 88; Ensembl ENSG00000077522
Protein	Alpha-actinin-2 — UniProt P35609
Locus	Chromosome 1q42–1q43
ICD-10	G71.0 (muscular dystrophy) — no MPD6-specific code
ICD-11	8C70 (primary/hereditary myopathy) — no MPD6-specific code
MeSH	Distal myopathy maps under "Distal Myopathies" / "Muscular Dystrophies"

Synonyms / alternative names: Distal myopathy 6 (MPD6); adult-onset autosomal dominant distal myopathy; **actininopathy**; ACTN2-related distal myopathy; ACTN2 myopathy (dominant, non-congenital).

Information source. The knowledge base for MPD6 is derived from **aggregated, disease-level resources** — case series and family cohorts reported in the primary literature (Savarese 2019; Iruzubieta 2025), plus OMIM/gene-level curation — rather than from individual patient EHR data. The largest cohort to date comprises 37 patients from 11 families [PMID: 39812845].

2. Etiology

Primary cause — genetic. MPD6 is a monogenic disorder caused by heterozygous **missense** (and in some kindreds protein-extending **frameshift**) variants in *ACTN2*. In the founding study, a unique *ACTN2* missense variant cosegregated with disease in all three Spanish families, while the Swedish family carried a different *ACTN2* missense variant: *"In all 3 Spanish families, we identified a unique missense variant in the ACTN2 gene cosegregating with the disease. The affected members of the Swedish family carry a different ACTN2 missense variant."* [PMID: 30900782].

Genetic risk factors. The causal variant in *ACTN2* is both necessary and (with incomplete, age-dependent penetrance) largely sufficient. There are **no established susceptibility loci or GWAS-identified common-variant risk factors** for this Mendelian disease. Certain recurrent variants behave as founder alleles in specific populations (see §9).

Environmental / lifestyle risk factors. None are established. As a dominantly inherited sarcomeric disease, onset and progression are driven by genotype and age, not by toxins, infections, or lifestyle. No occupational or exposure risk factors have been reported.

Protective factors. No genetic or environmental protective factors have been identified. Reduced penetrance in some carriers implies the existence of modifiers, but none have been mapped.

Gene–environment interactions. No specific gene–environment interactions have been documented for MPD6. Mechanical loading (eccentric contraction) is biologically plausible as an aggravating factor given the Z-disk's role as the structure most vulnerable to eccentric damage [PMID: 21536590], but this remains **inferred, not demonstrated**, for MPD6.

3. Phenotypes

MPD6 phenotypes are **clinical signs and physical manifestations** of skeletal muscle weakness, with supporting **laboratory/imaging abnormalities**. The phenotype is genotype-dependent and ranges from asymptomatic to limb-girdle weakness with facial involvement [PMID: 39812845].

Phenotype	Type	Onset	Severity / progression	Frequency	Suggested HPO
Distal lower-limb weakness (ankle dorsiflexion / foot drop)	Clinical sign	Adult	Progressive; initial site	Characteristic, near-universal at onset	HP:0009053 (distal lower limb muscle weakness); HP:0009027 (ankle weakness)
Asymmetric muscle weakness	Clinical sign	Adult	Progressive	Common	asymmetry of HP:0003701 (proximal muscle weakness)
Proximal spread / limb-girdle weakness	Clinical sign	Later in course	Progressive	Genotype-dependent	HP:0003701 (proximal muscle weakness)
Facial weakness	Clinical sign	Variable	Variable	Some genotypes	HP:0030319 (facial palsy/ weakness)
Gait disturbance / difficulty walking	Physical manifestation	Adult	Progressive	Common	HP:0001288 (gait disturbance)
Z-line disorganization / cores on biopsy	Lab (histology)	—	—	Present but non-specific	core myopathy features / HP:0003557 (increased fiber-diameter variability)
Cardiac involvement	Clinical sign	—	—	Not common	HP:0001638 (cardiomyopathy) — usually absent
Respiratory involvement	Clinical sign	—	—	Not common	HP:0002093 (respiratory)

Phenotype	Type	Onset	Severity / progression	Frequency	Suggested HPO
					insufficiency) — usually absent

The founding cohort described *"adult onset asymmetric distal muscle weakness with initial involvement of ankle dorsiflexion later progressing also to proximal limb muscles"* [PMID: 30900782]. The larger 37-patient cohort confirmed that *"the clinical phenotype ranged from asymptomatic to limb-girdle weakness and facial involvement and was depending on genotype. Cardiac and respiratory involvement were not common."* [PMID: 39812845].

Quality-of-life impact. Progressive foot drop and distal weakness impair gait, balance, and mobility, typically necessitating ankle-foot orthoses; proximal spread compromises stair climbing, rising from a chair, and daily activities. No disease-specific EQ-5D/SF-36/PROMIS data are published for MPD6; QoL impact is inferred to be moderate and progressive but non-life-limiting, given the usual sparing of cardiac and respiratory function.

4. Genetic / Molecular Information

Causal gene. ACTN2 (alpha-actinin-2), HGNC:164, NCBI Gene 88, chromosome 1q42–1q43, encoding UniProt **P35609**. ACTN2 is highly expressed in cardiac and skeletal muscle, is located in the sarcomere Z-disk, and links anti-parallel actin filaments while binding N-terminal titin — thereby contributing to sarcomere stability [PMID: 36116040].

Pathogenic variants.

Variant (protein)	cDNA	Type	Inheritance / phenotype	Population / note
p.Ile134Asn	—	Missense	Dominant distal/ LGMD	Belgian founder (shared haplotype) [PMID: 39812845]
p.Cys487Arg	—	Missense	Dominant distal/ LGMD	Basque-Spanish founder (shared haplotype) [PMID: 39812845]
p.Phe835Serfs*66	c.2504delT	Frameshift, C-terminal extension	Dominant, distal myopathy w/ multi-minicores	Aggregation-forming [PMID: 34170073]
p.Arg506Gly	c.1516A>G	Missense (biallelic)	Recessive (distinct entity)	Palestinian founder [PMID: 38311799]
c.1840-2A>T	—	Splice-site	Expands spectrum	[PMID: 36116040]

Variant classification (ACMG/AMP). Founder and cosegregating variants are classified pathogenic/likely pathogenic based on segregation and functional data. However, the *ACTN2* mutation-update review cautions that *"limited segregation and functional data are available to support the pathogenicity of most previously reported missense variants and clear-cut genotype-phenotype correlations are currently only demonstrated for some ACTN2-related myopathies."* [PMID: 36116040] — i.e., many *ACTN2* missense changes remain VUS.

Variant types: predominantly **missense** (dominant distal/LGMD and recessive forms) and **frameshift with C-terminal protein extension** (dominant, aggregation-prone). Congenital "Multiple structured Core Disease" (MsCD) arises from de novo missense and deletion variants [PMID: 30701273].

Allele frequency. Pathogenic *ACTN2* MPD6 variants are private/rare and essentially **absent or ultra-rare in gnomAD**, consistent with a highly penetrant dominant disease allele. (By contrast, the *ACTN3* R577X null polymorphism is common in ~20% of the general population — a different gene [PMID: 21536590].)

Somatic vs germline: **germline** (inherited, autosomal dominant). De novo germline events cause the congenital MsCD phenotype.

Functional consequences. The dominant distal disease acts through a **toxic gain-of-function** / **dominant-negative** mechanism (aggregation), not haploinsufficiency. Frameshift, protein-extending variants form alpha-actinin-2 aggregates [PMID: 39095936]; a specific frameshift *increased* alpha-actinin-2 expression and produced abnormal Z-line organization [PMID: 34170073]. For an MsCD missense variant, mutant protein localized correctly to the Z-line with unaltered levels, arguing **against** haploinsufficiency for that variant [PMID: 30701273].

Modifier genes. None specifically mapped for MPD6. Biologically, the paralog **ACTN3** modulates Z-disk composition and elastic properties (loss of alpha-actinin-3 up-regulates alpha-actinin-2 and changes Z-disk protein interactions) [PMID: 21536590], making *ACTN3* genotype a plausible but unproven modifier.

Epigenetic information / chromosomal abnormalities. No disease-specific DNA-methylation, histone-modification, or large-scale chromosomal abnormalities have been reported for MPD6. It is a point-mutation/small-indel disorder.

5. Environmental Information

Environmental factors: None established. MPD6 is a monogenic disease with no known toxic, radiation, pollution, or occupational contributors.

Lifestyle factors: None established as causal. Physical activity involving eccentric loading could theoretically stress a compromised Z-disk (inferred from Z-disk vulnerability to eccentric damage [PMID: 21536590]), but no clinical evidence links lifestyle to MPD6 onset or progression.

Infectious agents: Not applicable — MPD6 has no infectious etiology.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous ACTN2 variant** (missense, or frameshift causing a C-terminal protein extension) is inherited in the germline → **leads to** production of a structurally altered alpha-actinin-2 protein.
2. The altered protein **results in** reduced solubility and thermal destabilization of the alpha-actinin-2 molecule (demonstrated for actin-binding-domain substitutions) [PMID: 42481460] → **promotes** misfolding.
3. Misfolded/protein-extended alpha-actinin-2 **leads to** formation of **alpha-actinin-2 aggregates** within the muscle fiber (demonstrated for dominant protein-extending frameshift variants; not seen for recessive missense variants) [PMID: 39095936].
4. **Branch A (dominant distal / frameshift):** aggregation + increased alpha-actinin-2 expression → **abnormal Z-line organization** [PMID: 34170073].
5. **Branch B (congenital MsCD missense/deletion):** mutant protein incorporates at the Z-line and **results in** sarcomeric disorganization and **cores with Z-line defects** even without detectable aggregation [PMID: 30701273].
6. Z-line disorganization and core formation **result in** impaired sarcomere structural integrity and contractile force (impaired muscle force demonstrated in AAV-transduced mouse muscle; motor deficits in zebrafish) [PMID: 30701273].
7. Chronic sarcomeric dysfunction **leads to** progressive muscle fiber degeneration/loss (the defining feature shared with muscular dystrophies) [PMID: 33458580] — *inferred* for MPD6 from the dystrophic paradigm and biopsy findings rather than mechanistically dissected.
8. Fiber loss **manifests** first in distal lower-limb muscles (ankle dorsiflexors) — producing foot drop — then **spreads** proximally to limb-girdle muscles, with genotype-dependent severity [PMID: 30900782;

P 39812845

Inferred vs demonstrated: Steps 2–4 are experimentally demonstrated in cell and animal models; the coupling from aggregation to fiber loss (step 5) is **inferred**.

Mechanistic detail by category

- Protein dysfunction (central).** Alpha-actinin-2 is an **antiparallel homodimer** with three domains: an N-terminal **actin-binding domain (ABD)** comprising two calponin-homology (CH) domains; a central **rod** of four spectrin-like repeats mediating dimerization; and a C-terminal **calmodulin-like domain** with EF-hands. Disease variants **destabilize** the protein: *"All variants display reduced solubility, with actin-binding domain (ABD) substitutions showing pronounced thermal instability by differential scanning fluorimetry."* [PMID: 42481460], with SEC-MALS/SAXS revealing a strong propensity for aggregation. CH-domain mutations perturb F-actin binding and Z-disc localization/dynamics: *"both mutants have reduced F-actin binding affinity ... both mutations additionally affect Z-disc localization and dynamic behaviour."* [PMID: 27287556]. The dominant misfolding/aggregation is the unifying molecular lesion: *"dominant frameshift variants causing a protein extension do form alpha-actinin-2 aggregates. The results suggest that alpha-actinin-2 aggregation is the disease mechanism..."* [PMID: 39095936].
- Cellular processes.** Protein aggregation and proteostatic stress; abnormal Z-line assembly; myofibrillar disarray; fiber-size variability and eventual fiber loss. **GO terms:** GO:0030018 (Z disc), GO:0006457 (protein folding), GO:0030239 (myofibril assembly), GO:0006936 (muscle contraction).
- Molecular pathways / interactions.** Alpha-actinin-2 is a Z-disk hub binding actin, titin, and multiple Z-disk partners (ZASP/LDB3, myotilin, telethonin). Disruption of these interactions destabilizes the sarcomere. Beyond structure, the **spectrin-repeat domain binds and regulates ion channels** (see below) — relevant to *ACTN2* cardiac phenotypes but not the primary driver of the skeletal distal myopathy.
- Ion-channel / electrophysiological role (cardiac-relevant branch).** Alpha-actinin-2 interacts with the cardiac sodium channel Nav1.5: *"We identified alpha-actinin-2 as an interacting partner of the cytoplasmic loop connecting domains III and IV of Na(v)1.5 ... the spectrin-like repeat domain is essential for binding of alpha-actinin-2 to Na(v)1.5."* [PMID: 19943616], and *"the interaction with alpha-actinin-2 increases sodium channel density without changing their gating properties"* [PMID: 19943616]. This explains why some *ACTN2* variants present as idiopathic ventricular fibrillation [PMID: 35975100] — a phenotype distinct from MPD6, in which cardiac involvement is uncommon.
- Metabolic changes / immune involvement:** No primary metabolic or autoimmune mechanism; MPD6 is a structural sarcomeric proteinopathy, not an inflammatory or metabolic myopathy.

- **Tissue damage mechanism:** Mechanical failure of the Z-disk under contractile load with progressive myofiber degeneration; the Z-disk is the sarcomeric structure most vulnerable to (eccentric) mechanical damage [PMID: 21536590].
- **Transcriptomics/proteomics:** Variant-specific increases in alpha-actinin-2 protein levels have been observed for a frameshift allele [PMID: 34170073]; no comprehensive MPD6 muscle transcriptome/proteome dataset is yet available.

Upstream vs downstream: Upstream = the *ACTN2* mutation and consequent protein misfolding/aggregation; downstream = Z-line disorganization → contractile deficit → fiber loss → clinical weakness. **Cell type:** skeletal muscle fiber (**CL:0000188**). **Biological process GO:** GO:0030239 (myofibril assembly), GO:0006936 (muscle contraction).

7. Anatomical Structures Affected

Organ / system level. Primary organ affected: **skeletal muscle** (musculoskeletal / muscular system). In the dominant distal form, cardiac and respiratory systems are **usually spared** [PMID: 39812845]. (Cardiac muscle is affected in the separate *ACTN2*-cardiomyopathy spectrum, not in MPD6.)

Distribution (muscle MRI signature). In the 37-patient cohort, *"the most affected muscles were the glutei minor, glutei medius, hamstrings, tibialis anterior, and soleus"* [PMID: 39812845]. A machine-learning model found the most differentiating features were involvement of **tibialis anterior and gluteus medius with preservation of quadratus femoris** [PMID: 39812845]. The recessive form shows selective hamstrings/adductors (thigh) and anterior tibial group/soleus (lower leg) [PMID: 38311799].

Level	Structure	Suggested UBERON / CL / GO
Organ/ system	Skeletal muscle (musculoskeletal system)	UBERON:0001134 (skeletal muscle tissue)
Muscle group	Tibialis anterior	UBERON:0001385
Muscle group	Soleus	UBERON:0011906
Muscle group	Gluteus medius/minimus	UBERON:0001500 (gluteal muscle)
Muscle group	Hamstrings	UBERON:0011907 (biceps femoris)
Cell	Skeletal muscle fiber	CL:0000188 (skeletal muscle cell)
Subcellular	Sarcomere Z-disk	GO:0030018 (Z disc)
Subcellular	Myofibril / sarcomere	GO:0030016 (myofibril), GO:0030017 (sarcomere)

Lateralization: Characteristically **asymmetric** [PMID: 30900782; 38311799].

8. Temporal Development

Onset. Adult-onset (the "adult-onset" descriptor is part of the disease name). Onset pattern is **insidious/chronic**, beginning with asymmetric ankle-dorsiflexion weakness (foot drop) [PMID: 30900782]. (De novo *ACTN2* variants can cause a congenital core myopathy — MsCD — a distinct, earlier-onset presentation [PMID: 30701273].)

Progression. Slowly progressive. Weakness begins distally in the lower limbs and later spreads proximally to limb-girdle muscles [PMID: 30900782]. Severity is genotype-dependent, spanning asymptomatic carriers to limb-girdle weakness with facial involvement [PMID: 39812845].

Course / duration. **Chronic, lifelong**, progressive. No relapsing-remitting or episodic pattern; no spontaneous remission. Because cardiac and respiratory systems are usually spared, the course is generally not life-limiting.

Critical periods / intervention windows. No pharmacologic window is defined (no disease-modifying therapy exists). The practical intervention window is symptomatic — orthotic and rehabilitative management once distal weakness begins.

9. Inheritance and Population

Epidemiology. MPD6 is **ultra-rare**. No formal prevalence/incidence figures are established; knowledge derives from a small number of families. The largest published series comprises **37 patients from 11 families** (Spanish and Belgian) [PMID: 39812845]; the founding report described 4 families [PMID: 30900782].

Inheritance. **Autosomal dominant** (heterozygous *ACTN2* variant), with **incomplete, age-dependent penetrance** and **variable, genotype-dependent expressivity** [PMID: 39812845]. A biologically and clinically **distinct autosomal recessive** *ACTN2* myopathy exists (biallelic p.Arg506Gly) and must not be conflated with MPD6 [PMID: 38311799].

Penetrance / expressivity. Penetrance is incomplete and age-dependent (asymptomatic adult carriers occur); expressivity ranges from asymptomatic to limb-girdle weakness with facial involvement depending on the specific variant [PMID: 39812845].

Anticipation / mosaicism: No evidence of genetic anticipation (not a repeat-expansion disorder). Germline mosaicism has not been specifically reported.

Founder effects. - **p.Ile134Asn** — Belgian founder allele (unique shared haplotype indicating common ancestry) [PMID: 39812845]. - **p.Cys487Arg** — Basque-Spanish founder allele (unique shared haplotype) [PMID: 39812845]. - (Recessive) **p.Arg506Gly** — Palestinian founder, confirmed by haplotype analysis: *"all patients report Palestinian ethnicity, suggesting a possible founder ACTN2 variant, which was confirmed through haplotype analysis in two families"* [PMID: 38311799].

"Belgian families carrying the p.Ile134Asn variant and Basque-Spanish families carrying the p.Cys487Arg variant each showed unique haplotypes supporting respective common ancestry." [PMID: 39812845].

Consanguinity / carrier frequency. Consanguinity is relevant to the recessive form (Palestinian founder) but not to dominant MPD6. Dominant pathogenic alleles are private/ultra-rare in gnomAD; no meaningful carrier frequency applies to a dominant disease.

Population demographics. Reported families are of Spanish (including Basque), Belgian, and Swedish origin (dominant form) [PMID: 30900782; [P 39812845](#)], and Palestinian (recessive form) [PMID: 38311799]. No strong sex bias is reported (autosomal inheritance). Age distribution reflects adult onset.

10. Diagnostics

Genetic testing (definitive). Diagnosis rests on identification of a pathogenic *ACTN2* variant. The recommended approach is **next-generation sequencing** — a distal-myopathy/neuromuscular **gene panel** (which should include *ACTN2* alongside the ~20 dominant distal-myopathy genes such as *DNM2*, *MATR3*, *VCP*, *MYOT*, *LDB3/ZASP*, *FLNC*, *TIA1*, *HNRNPA1*, *DES*, *MYH7*, *TTN*) [PMID: 33458580], or **whole-exome sequencing** with segregation analysis. Single-gene *ACTN2* testing is appropriate when the MRI/clinical picture is characteristic or a family variant is known. Because many *ACTN2* missense variants remain VUS, **segregation and functional evidence are often needed** to confirm pathogenicity [PMID: 36116040].

Muscle MRI (key adjunct). A characteristic selective-involvement pattern supports the diagnosis and directs biopsy site: predominant involvement of **tibialis anterior, soleus, gluteus medius/minimus, and hamstrings**, with relative preservation of quadratus femoris [PMID: 39812845]. Muscle MRI is broadly established as crucial for myopathy diagnosis, biopsy-site selection, and activity assessment [PMID: 41066296].

Muscle biopsy / histopathology. Findings are **non-specific** (myopathic changes; variability in fiber diameter) but may show **Z-line disorganization, cores/multi-minicores, and alpha-actinin-2 aggregates/abnormal Z-line organization** [PMID: 34170073; [P 30701273](#)].

Immunohistochemistry for alpha-actinin-2 can reveal aggregation.

Laboratory tests. Serum **creatine kinase (CK)** is typically normal or only mildly elevated (as in many distal myopathies). No specific circulating biomarker exists.

Electrophysiology. EMG shows a myopathic pattern; nerve conduction studies are normal (distinguishing from neuropathy).

Clinical criteria / differential diagnosis. No formal consensus criteria exist; diagnosis integrates adult-onset asymmetric distal (foot-drop) weakness, characteristic MRI, myopathic EMG, supportive biopsy, and a confirmed *ACTN2* variant. **Differential diagnosis** includes other distal myopathies (e.g., Welanders, Markesbery-Griggs/ZASP, Udd/titinopathy, GNE myopathy, dysferlinopathy, myofibrillar myopathies) [PMID: 33458580], **inclusion body myositis** (asymmetric distal/proximal weakness, normal CK possible) [PMID: 37038366], and acquired causes of distal/asymmetric weakness such as vasculitic or inflammatory myopathy [PMID: 37900606; 33863518].

P 33863518

Omics-based diagnostics: Not routinely used; RNA-seq/proteomics are research tools.

Screening. No population or newborn screening exists (ultra-rare, adult-onset, dominant).

Cascade genetic testing of at-risk relatives is appropriate once a familial variant is identified.

11. Outcome / Prognosis

Survival / mortality. Generally **not life-limiting** in the dominant distal form, because "*Cardiac and respiratory involvement were not common.*" [PMID: 39812845]. No disease-specific mortality data are published.

Morbidity / function. The main burden is progressive **mobility impairment** from foot drop and distal-to-proximal weakness, often requiring ankle-foot orthoses and, with proximal spread, assistance for stairs and transfers. Facial weakness occurs with some genotypes [PMID: 39812845]. No formal disability/QoL instrument data (EQ-5D, SF-36, PROMIS) are published for MPD6.

Disease course / complications. Slowly progressive weakness; complications are secondary to reduced mobility (falls, deconditioning). Cardiomyopathy and respiratory failure are **uncommon** in dominant MPD6 [PMID: 39812845] — a key prognostic distinction from *ACTN2*-cardiomyopathy.

Recovery potential. None — progressive and irreversible; management is supportive, not curative.

Prognostic factors. Genotype is the principal prognostic determinant (asymptomatic → limb-girdle severity is genotype-dependent) [PMID: 39812845]. Aggregation-forming (protein-extending) variants are mechanistically more clearly pathogenic [PMID: 39095936]. No validated prognostic biomarkers exist.

12. Treatment

There is no disease-modifying or curative therapy for MPD6. Management is **supportive and rehabilitative**.

- **Supportive / rehabilitative care (mainstay):** Ankle-foot orthoses (AFOs) for foot drop; physiotherapy to maintain range of motion and strength; occupational therapy and assistive/mobility devices as proximal weakness develops; fall-prevention. (NCIT: *Physical Therapy* C15917; *Orthotic Device* C50094; *Rehabilitation Therapy* C15326.)
- **Pharmacotherapy:** None disease-specific. No approved drug targets alpha-actinin-2 aggregation.
- **Advanced therapeutics (none approved; conceptual):** For a dominant toxic gain-of-function/aggregation mechanism, **allele-selective silencing** (antisense oligonucleotides or siRNA targeting the mutant *ACTN2* allele) is a rational future strategy — but **no clinical trials or NCT identifiers exist** for MPD6.
- **Surgical:** None specific; orthopedic interventions (e.g., for contractures) only as needed.
- **Cardiac/respiratory monitoring:** Given that cardiac/respiratory involvement is uncommon [PMID: 39812845] but *ACTN2* is a recognized cardiomyopathy gene, **baseline cardiac evaluation** (ECG/echocardiography) is prudent, individualized to genotype and family history.
- **Pharmacogenomics / personalized medicine:** Not applicable currently; care is genotype-informed mainly for counseling and prognosis.

Treatment outcomes / adverse events: Rehabilitative measures improve function and safety without disease-specific pharmacologic adverse events; no response-rate data from trials exist.

13. Prevention

- **Primary prevention:** Not applicable to an inherited dominant disease beyond **reproductive options**. Preimplantation genetic testing (PGT) or prenatal diagnosis is possible when a familial pathogenic *ACTN2* variant is known.
 - **Secondary prevention: Cascade genetic testing** of at-risk relatives enables early identification, counseling, and baseline cardiac assessment.
 - **Tertiary prevention:** Rehabilitation, orthotics, fall-prevention, and contracture prevention to limit disability once symptomatic.
 - **Genetic counseling (central):** Essential — 50% transmission risk per offspring for a heterozygous carrier; counseling must address incomplete/age-dependent penetrance and variable, genotype-dependent expressivity [PMID: 39812845]. For the distinct recessive *ACTN2* myopathy, carrier counseling in founder populations (e.g., Palestinian for p.Arg506Gly) applies [PMID: 38311799].
 - **Immunization / public health / environmental interventions:** Not applicable (non-infectious, non-environmental Mendelian disease).
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14. Other Species / Natural Disease

- **Taxonomy / orthologs.** *ACTN2* is evolutionarily conserved. Model orthologs: **mouse** *Actn2* (*Mus musculus*, NCBI Taxon 10090; NCBI Gene 11472) and **zebrafish** *actn2* (*Danio rerio*, NCBI Taxon 7955). Both have been used to model *ACTN2* disease [PMID: 30701273].
 - **Natural disease in animals.** No well-characterized naturally occurring *ACTN2* distal myopathy is documented in companion animals or wildlife (no established OMIA entry for a spontaneous animal actininopathy).
 - **Comparative biology.** The conserved Z-disk role of alpha-actinin-2 across vertebrates underlies successful cross-species modeling; wild-type alpha-actinin-2 expression does not cause anomalies, whereas mutant protein produces sarcomeric disorganization in both zebrafish and mouse — demonstrating conservation of the disease mechanism [PMID: 30701273].
 - **Transmission / zoonosis:** Not applicable.
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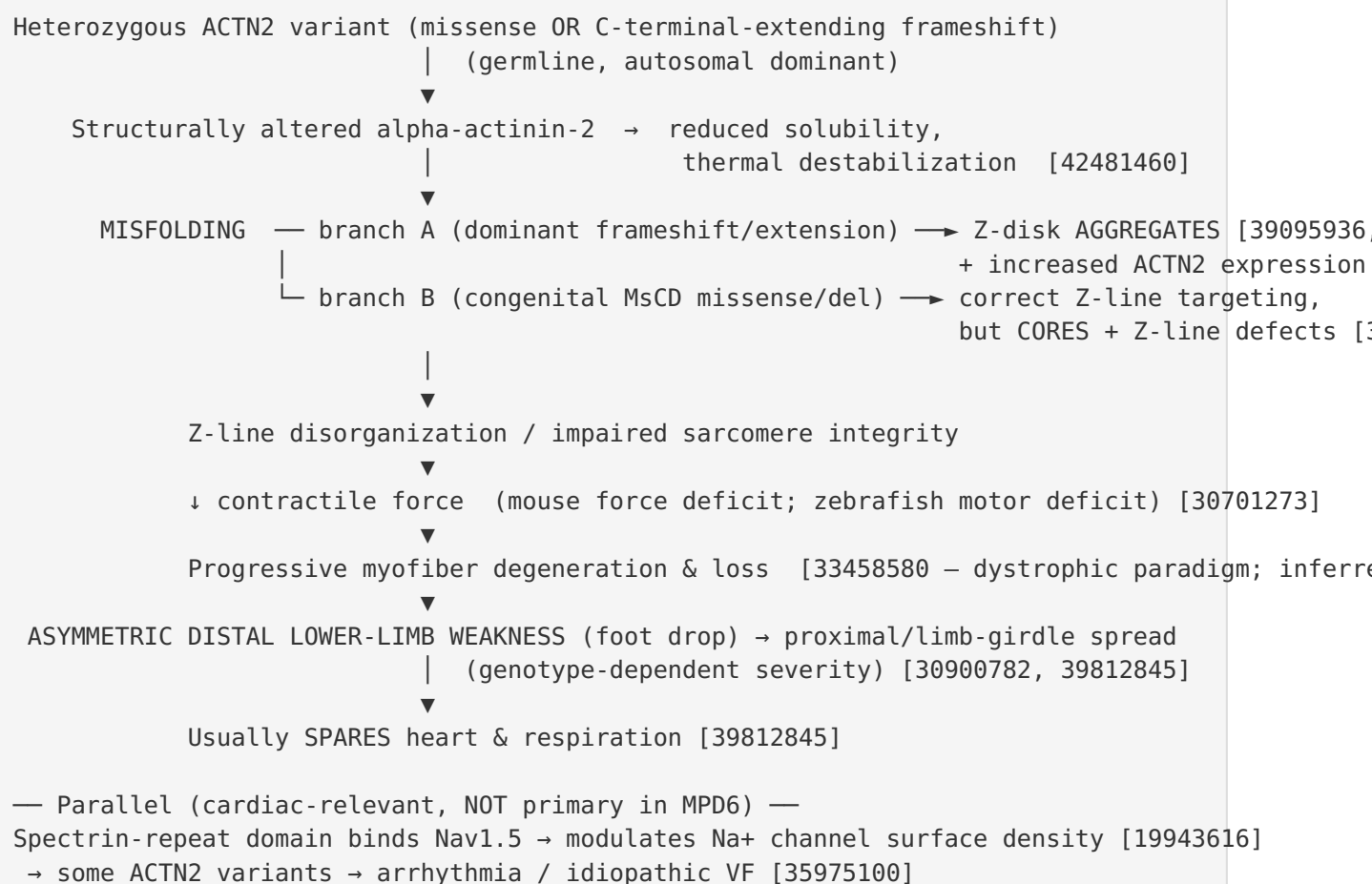
15. Model Organisms

Model	Type	Genetic approach	Key phenotype recapitulated	Reference
Zebrafish (<i>Danio rerio</i>)	Vertebrate, in vivo	Exogenous expression of mutant alpha-actinin-2	Motor deficits; sarcomeric disorganization	[PMID: 30701273]
Mouse (<i>Mus musculus</i>)	Mammalian, in vivo	AAV-mediated expression of mutant <i>ACTN2</i> in muscle	Impaired muscle force; cores and Z-line defects ; sarcomeric disorganization	[PMID: 30701273]
C2C12 myoblasts/myotubes	Cellular, in vitro	Expression of <i>ACTN2</i> variants	Alpha-actinin-2 aggregates (dominant frameshift); no aggregates (recessive missense)	[PMID: 39095936]
Cultured myotubes	In vitro	MsCD missense variant	Correct Z-line localization, unaltered levels (argues against haploinsufficiency)	[PMID: 30701273]

Phenotype recapitulation. Animal models faithfully reproduce the human structural pathology: *"Modelling of the disease in zebrafish and mice by exogenous expression of mutated alpha-actinin-2 recapitulated the abnormal muscle function and structure seen in the patients. Motor deficits were noted in zebrafish, and muscle force was impaired in isolated muscles from AAV-transduced mice."* [PMID: 30701273], and *"In both models, sarcomeric disorganization was evident, while expression of wild-type alpha-actinin-2 did not result in muscle anomalies. The murine muscles injected with mutant ACTN2 displayed cores and Z-line defects."* [PMID: 30701273]. The C2C12 model recapitulates the dominant aggregation mechanism and distinguishes it from recessive variants [PMID: 39095936].

Limitations. Existing models use **exogenous/overexpression** of mutant protein (zebrafish, AAV-mouse) rather than physiological knock-in at endogenous levels; they capture structural pathology and force deficits but not the slow adult-onset, distal-predominant progression seen in patients. No germline knock-in mouse recapitulating adult-onset distal MPD6 has been reported.

Mechanistic Model / Interpretation



Synthesis. MPD6 is best understood as a **dominant sarcomeric proteinopathy of the Z-disk**. The unifying lesion is a destabilized alpha-actinin-2 that misfolds and — for dominant protein-extending alleles — aggregates, poisoning Z-disk assembly (a dominant-negative/toxic gain-of-function effect rather than simple loss of function). This distinguishes MPD6 both from the **recessive** *ACTN2* myopathy (missense, non-aggregating, distinct phenotype and Palestinian founder) [PMID: 38311799] and from ***ACTN2* cardiomyopathy/arrhythmia**, where the protein's separate ion-channel-scaffolding role via its spectrin-repeat domain becomes pathogenically relevant [PMID: 19943616; 35975100].

P 35975100

The single gene *ACTN2* thus produces a spectrum — dominant distal MPD6, congenital MsCD, recessive myopathy, and cardiomyopathy — whose divergent phenotypes track with variant type, domain location, aggregation propensity, and zygosity.

Evidence Base

PMID	Study	Role in this report
30900782	<i>Actininopathy: A new muscular dystrophy caused by ACTN2 dominant mutations</i> (Savarese et al.)	Defining paper — dominant ACTN2 missense variants, adult-onset asymmetric distal phenotype, disease concept
39812845	<i>Clinical and imaging spectrum of non-congenital dominant ACTN2 myopathy</i> (Iruzubieta et al.)	Largest cohort (37 pts/11 families) — genotype-dependent phenotype, MRI signature, founder alleles, cardiac/respiratory sparing
39095936	<i>Protein-extending ACTN2 frameshift variants cause variable myopathy phenotypes by protein aggregation</i>	Establishes aggregation as the dominant mechanism (frameshift aggregates; recessive missense do not)
34170073	<i>A novel frameshift ACTN2 variant causes a rare adult-onset distal myopathy with multi-minicores</i>	Independent family: increased ACTN2, abnormal Z-line, aggregate formation
30701273	<i>ACTN2 mutations cause "Multiple structured Core Disease" (MsCD)</i> (Lornage et al.)	Zebrafish + mouse models ; cores/Z-line defects; argues against haploinsufficiency for MsCD variant
42481460	<i>Comprehensive biophysical/structural profiling of alpha-actinin-2 variants</i> (Nouredine et al.)	Variant destabilization, reduced solubility, aggregation propensity
27287556	<i>HCM mutations in the CH domain of ACTN2 affect actin binding and Z-disc incorporation</i>	CH-domain variants alter actin binding and Z-disc dynamics
19943616	<i>Cell membrane expression of Nav1.5 is modulated by alpha-actinin-2 interaction</i> (Ziane et al.)	Spectrin-repeat/Nav1.5 interaction → cardiac electrophysiology branch
35975100	<i>Novel ACTN2 missense variant associated with idiopathic VF</i>	ACTN2 arrhythmia phenotype (distinct from MPD6)
38311799	<i>Recurring homozygous ACTN2 variant (p.Arg506Gly) causes a recessive myopathy</i>	Distinct recessive entity ; Palestinian founder
36116040	<i>Mutation update for the ACTN2 gene</i>	Domain/localization biology; caution on VUS and genotype–phenotype correlations

PMID	Study	Role in this report
33458580	<i>Panorama of the distal myopathies</i>	Nosology; <i>ACTN2</i> among ~20 dominant distal-myopathy genes; dystrophic paradigm
21536590	<i>α-actinin-3 deficiency and contraction-induced damage</i>	Z-disk vulnerability to eccentric damage; <i>ACTN2/ACTN3</i> Z-disk biology (modifier plausibility)
41066296	<i>Myopathies: Radiologist's essential tips</i>	Role of muscle MRI in myopathy diagnosis/biopsy targeting

Evidence source types: human clinical/genetic (30900782, 39812845, 34170073, 38311799, 35975100, 36116040); in vivo model organism (30701273 — zebrafish/mouse); in vitro cell/biophysical (39095936, 42481460, 27287556, 19943616); review/nosology (33458580, 41066296, 21536590).

Limitations and Knowledge Gaps

- 1. Small evidence base.** The entire dominant-disease literature rests on a handful of families (largest series 37 patients) [PMID: 39812845]; prevalence, incidence, sex ratio, and natural-history metrics are not formally established.
- 2. VUS burden.** Many *ACTN2* missense variants lack segregation/functional confirmation; genotype–phenotype correlations are demonstrated for only some variants [PMID: 36116040].
- 3. Mechanistic gap (aggregation → fiber loss).** The step from Z-disk aggregation to progressive myofiber degeneration is inferred from the dystrophic paradigm, not dissected in MPD6 muscle.
- 4. Model limitations.** Existing zebrafish/mouse models rely on overexpression rather than endogenous knock-in and do not reproduce the slow, adult-onset, distal progression [PMID: 30701273].
- 5. No omics datasets.** No published MPD6-specific transcriptomic, proteomic, or single-cell datasets exist.
- 6. No QoL / outcome instruments.** No EQ-5D/SF-36/PROMIS or validated functional-outcome data for MPD6.

7. **No therapeutics.** No disease-modifying treatment, biomarker, or clinical trial (no NCT) exists.

Proposed Follow-up Experiments / Actions

1. **Generate an endogenous knock-in mouse** for a recurrent dominant allele (e.g., p.Cys487Arg or a protein-extending frameshift) to model adult-onset distal progression and test disease-modifying strategies.
 2. **Allele-selective silencing proof-of-concept.** Test mutant-allele-specific ASO/siRNA in patient iPSC-derived myotubes and knock-in models, given the dominant aggregation mechanism [PMID: 39095936].
 3. **Aggregation-focused biomarkers.** Quantify alpha-actinin-2 aggregate load and proteostatic-stress markers in patient biopsies as candidate progression biomarkers.
 4. **Multi-omics of patient muscle.** Bulk/single-nucleus RNA-seq and proteomics of affected vs preserved muscles (guided by the MRI signature [PMID: 39812845]) to map downstream degeneration pathways.
 5. **Prospective natural-history registry** with standardized MRI (tibialis anterior/soleus/gluteus medius scoring) and functional outcomes to define progression rate and penetrance curves.
 6. **Systematic variant reclassification.** Combine biophysical destabilization/aggregation assays [PMID: 42481460] with segregation data to resolve *ACTN2* VUS.
 7. **Cardiac surveillance study.** Prospectively assess subclinical cardiac/electrophysiologic involvement across MPD6 genotypes, given alpha-actinin-2's Nav1.5-scaffolding role [PMID: 19943616].
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Ontology Term Quick-Reference

- **Disease:** MONDO:0014138; OMIM:615197
- **Gene/protein:** ACTN2 (HGNC:164, NCBI Gene 88); UniProt P35609

- **Phenotype (HPO):** HP:0009053 (distal lower limb muscle weakness), HP:0003701 (proximal muscle weakness), HP:0001288 (gait disturbance), HP:0030319 (facial weakness), HP:0001638 (cardiomyopathy — usually absent)
 - **Cell (CL):** CL:0000188 (skeletal muscle cell / myofiber)
 - **Anatomy (UBERON):** UBERON:0001134 (skeletal muscle tissue), UBERON:0001385 (tibialis anterior), UBERON:0011906 (soleus), UBERON:0001500 (gluteal muscle)
 - **Subcellular / process (GO):** GO:0030018 (Z disc), GO:0030016 (myofibril), GO:0030239 (myofibril assembly), GO:0006457 (protein folding), GO:0006936 (muscle contraction)
 - **Treatment (NCIT):** C15917 (Physical Therapy), C50094 (Orthotic Device), C15326 (Rehabilitation Therapy)
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Report compiled from 5 completed investigation iterations, 30 reviewed papers, and 8 confirmed findings. All quoted statements are verbatim from the cited PubMed abstracts.

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